

Google Pixel 3a XL (bonito)

PM660 USB-C Dual-Role Engineering Validation

postmarketOS / Linux 7.0.10-sdm670
Known-good build: 7.0.10-sdm670-gaac5fb159b67

Engineering test report, change rationale, validation evidence, deployment notes, and patch-series inventory

Device	Google Pixel 3a XL (bonito), Qualcomm SDM670 / PM660
OS	postmarketOS v26.06 environment
Kernel	7.0.10-sdm670-gaac5fb159b67
Branch	bonito-pm660-typec-test
Final Git HEAD	aac5fb159b67d616e0aa683dd06324a5cc517126

1. Executive Summary

USB-C dual-role operation on the Pixel 3a XL was brought to a functional, repeatable state by adding PM660 VBUS source control, implementing the PM660 Type-C backend for TCPM, and correcting the timing and notification semantics between PM660, the regulator framework, and Linux TCPM.

The final tested build successfully operates in both major USB-C role combinations: SOURCE + HOST when an OTG device is attached, and SINK + DEVICE when connected to a computer. Host mode was validated with a Kingston DataTraveler and a powered USB dock. Device mode was validated with the DWC3 UDC in configured state, high-speed gadget enumeration, an active NCM usb0 link, and simultaneous charging.

2. Original Failure Modes

- Initial PM660 Type-C work could probe and expose TCPM, but VBUS source control was incomplete.
- An early implementation called `tcpm_vbus_change()` from `pm660_typec_set_vbus()` before `tcpm_port` was guaranteed to be valid.
- That produced a kernel NULL-pointer Oops during TCPM initialization/reset.

Observed failure stack:

```
tcpm_vbus_change
pm660_typec_set_vbus
tcpm_reset_port
tcpm_init
tcpm_register_port
qcom_pmic_typec_probe
```

- After removing the unsafe notification, host attach could reach `SRC_ATTACHED` and briefly request host mode, but TCPM never learned that source VBUS had become present.
- TCPM therefore hit its 480 ms `PD_T_PS_SOURCE_ON` fallback and returned to `SRC_UNATTACHED`.

3. Root Cause

The core issue was a mismatch between PM660 source-VBUS behavior and the event model expected by TCPM. For source attach, TCPM calls `set_vbus(true)`, then expects a VBUS-present event to advance `SRC_ATTACHED` to `SRC_STARTUP`. PM660's hardware status did not always immediately reflect sourced VBUS, so relying only on the PMIC VBUS status bit was insufficient.

Failure sequence before final fix:

```
SRC_ATTACH_WAIT
-> SRC_ATTACHED
-> vbus:=1
-> USB_ROLE_HOST requested
-> no VBUS-on notification
-> 480 ms timeout
-> SRC_UNATTACHED
```

The final approach makes the driver's software `vbus_enabled` state authoritative for self-sourced VBUS, while still tracking real PM660 VBUS status changes in the Type-C IRQ path.

4. Implemented Changes

4.1 PM660 SMB2 VBUS regulator

`qcom_smbx` was extended with a fixed 5 V VBUS regulator backed by the PM660 SMB2 OTG boost path. The regulator implements enable, disable, and `is_enabled` operations.

- USB input is suspended before enabling OTG.
- The PM660 buck/boost workaround is configured.
- OTG startup current is ramped through 250 mA, 500 mA, 1 A, and 1.5 A while waiting for BOOST_SOFTSTART_DONE.
- On success, normal source current is restored; disable reverses the sequence and resumes USB input.

4.2 PM660 Type-C backend

A PM660-specific backend was added to the Qualcomm PMIC TCPM driver. It supplies get_vbus, set_vbus, get_cc, set_cc, polarity, VCONN, DRP toggling, and PM660 port start/stop operations.

- PM660 CC status is decoded into standard TCPM CC states.
- Try.SNK behavior is retained for bonito's sink-preferred policy.
- The aggregate Type-C interrupt always reports CC changes to TCPM.
- VBUS events are reported only on the first observed hardware VBUS state or when that state actually changes.

4.3 Safe VBUS notification from set_vbus()

The final set_vbus() implementation records whether the regulator state actually changed. After releasing the PM660 driver mutex, it notifies TCPM only if the state changed and tcpm_port has already been established.

```
Final behavior:
regulator_enable()/regulator_disable()
-> update port->vbus_enabled
-> unlock
-> if (changed && port->tcpm_port)
    tcpm_vbus_change(port->tcpm_port)
```

This avoids the earlier NULL-pointer initialization crash while still delivering the event TCPM needs to advance from SRC_ATTACHED to SRC_STARTUP.

5. Patch Series / Commit Inventory

Commit	Subject
f2d581bc5607	usb: typec: qcom: make PMIC port probe selectable
cc696116fc6d	usb: typec: qcom: add private port backend data
620f1b31c9d5	power: supply: qcom_smbx: add PM660 VBUS regulator
fc78301d6d8f	usb: typec: qcom: add PM660 Type-C port support
a202d57eaeac	dt-bindings: usb: qcom,pmic-typec: add PM660
655704b4bf15	dt-bindings: power: supply: qcom,pmi8998-charger: add VBUS regulator
afe6d1be1b53	arm64: dts: qcom: sdm670-google: enable PM660 USB-C dual-role
5d6164bc79ce	power: supply: qcom_smbx: define USB input suspend register
f5f301628e8f	usb: typec: qcom: pm660: avoid VBUS notification from set_vbus
b26caf853a77	power: supply: qcom_smbx: fix PM660 VBUS node lookup
3505a17f0bd3	usb: typec: qcom: pm660: only report CC changes from Type-C IRQ
64fb5a59470c	usb: typec: qcom: pm660: track VBUS changes in Type-C IRQ
aac5fb159b67	usb: typec: qcom: pm660: notify TCPM after VBUS changes

Patch base: 06f3bc759bf5 (tag: sdm670-v7.0.10). Final HEAD: aac5fb159b67d616e0aa683dd06324a5cc517126.

6. Runtime Validation - Host / Source

Test topology: Pixel 3a XL -> USB-C OTG adapter -> Kingston DataTraveler.

Signal	Observed value
Type-C power_role	[source] sink
Type-C data_role	[host] device
USB role switch	host
pm660-vbus state	enabled
pm660-vbus users	1
PM660 status registers	130b: 00, 130c: 28, 130e: 91
USB device	0951:1666 Kingston Technology DataTraveler
Block device	/dev/sda, partitions sda1 and sda2
Capacity reported	241,660,916 x 512-byte sectors (~124 GB / 115 GiB)

The TCPM state trace showed the critical successful transition:

```
CC1: 0 -> 2
TOGGLING -> SRC_ATTACH_WAIT
SRC_ATTACH_WAIT -> SRC_ATTACHED
vbus:=1
Requesting mux state 1, usb-role 1
VBUS on
SRC_ATTACHED -> SRC_STARTUP
```

7. Runtime Validation - Detach

```
Observed detach trace:
CC1: 2 -> 0 ... disconnected
SRC_READY -> SNK_UNATTACHED
Start toggling
VBUS off
VBUS VSAFE0V
```

- USB role returned to device.
- pm660-vbus returned to disabled.
- VBUS users returned to 0.
- Raw PM660 status returned to 130b: 00, 130c: 20, 130e: 00 in the captured detached state.

8. Runtime Validation - Peripheral / Sink / Charging

Test topology: MacBook -> USB cable -> Pixel 3a XL.

Signal	Observed value
Type-C power_role	source [sink]
Type-C data_role	host [device]
USB role switch	device
power_operation_mode	usb_power_delivery
preferred_role	sink
UDC state	configured
UDC current speed	high-speed
UDC maximum speed	high-speed
Gadget UDC	a600000.usb
usb0 carrier	1
usb0 operstate	up
PM660 registers	130b: 20, 130c: 00, 130e: 53

8.1 Charging evidence

Power-supply property	Observed value
pm660-charger status	Charging
online	1
health	Good
USB type	SDP
input voltage	5,039,062 μ V (~5.039 V)
input current	483,471 μ A (~483 mA)
current_max	500,000 μ A (500 mA)
battery status	Charging
battery capacity	23%
battery voltage	~3.76-3.77 V
sampled battery current	~62,988 to 179,199 μ A positive charge current

9. Functional Validation Matrix

Function	Result
Kernel boots normally	PASS
PM660 Type-C probe / TCPM registration	PASS
Dual-role / DRP behavior	PASS
SOURCE + HOST	PASS
5 V VBUS sourcing	PASS
xHCI host controller	PASS
USB mass storage	PASS
USB hub / powered dock enumeration	PASS
Host detach and VBUS shutdown	PASS
SINK + DEVICE	PASS
DWC3 UDC configured	PASS
ConfigFS NCM gadget	PASS
High-speed peripheral enumeration	PASS
USB charging from computer	PASS
Previous tcpm_vbus_change NULL Oops	RESOLVED

10. Known Quirks / Follow-up

- After some detach sequences, Type-C class power_role/data_role may retain the last active source/host presentation even though TCPM has returned to SNK_UNATTACHED, VBUS is off, and the actual USB role switch is device.
- A direct C-to-C ESP32 test was not a reliable host validation path; the Pixel repeatedly worked with a standards-compliant OTG adapter, flash drive, and dock. This points to the peripheral/cable CC advertisement rather than the Pixel host implementation.
- A separate touchscreen/session behavior was observed during testing and was intentionally left outside the scope of the USB-C patch set.

11. Reproduction / Proof Commands

```
# Build identity
uname -a
cat /usr/share/kernel/postmarketos-qcom-sdm670/kernel.release
modinfo -k "$(uname -r)" qcom_pmic_tcpm | grep -E '^(filename|name|vermagic):'

# Host/source state
cat /sys/class/typec/port0/power_role
cat /sys/class/typec/port0/data_role
```

```

cat /sys/class/usb_role/a600000.usb-role-switch/role
lsusb

# Find PM660 VBUS regulator
VBUS="$(for d in /sys/class/regulator/regulator.*; do
[ -f "$d/name" ] || continue
[ "$(cat "$d/name")" = "pm660-vbus" ] && echo "$d"
done)"
cat "$VBUS/state"
cat "$VBUS/num_users"

# PM660 raw Type-C status
sudo grep -Ei '^(130b|130c|130e):' /sys/kernel/debug/regmap/0-00/registers

# Device/sink state
UDC=/sys/class/udc/a600000.usb
cat "$UDC/state"
cat "$UDC/current_speed"
cat /sys/class/net/usb0/carrier
cat /sys/class/net/usb0/operstate

# Charging
for f in status online health usb_type voltage_now current_now current_max; do
[ -f "/sys/class/power_supply/pm660-charger/$f" ] &&
echo "$f: $(cat /sys/class/power_supply/pm660-charger/$f)"
done

```

12. Deployment and Recovery Notes

The known-good boot image was initially tested with fastboot boot, which RAM-boots the image and does not persist it. For permanent deployment, flash the tested image to the currently active boot slot only after verifying the slot.

```

# On the Mac, while phone is in fastboot:
./fastboot devices
./fastboot getvar current-slot

# If the result is 'b' (the tested device previously used slot b):
./fastboot flash boot_b ~/Desktop/bonito-pm660-vbus-notify.img

# Keep the same active slot and reboot:
./fastboot set_active b
./fastboot reboot

# After boot:
uname -r
# Expected:
# 7.0.10-sdm670-gaac5fb159b67

```

Do not flash boot_a merely for symmetry. On the tested phone, slot a had previously been marked unbootable while slot b was the successful slot. Preserve the original postmarketOS boot image on the Mac as the rollback artifact. If the permanent image fails, return to fastboot and either RAM-boot the known stock/postmarketOS image or re-flash boot_b with the saved working predecessor.

13. Upstream / Sharing Recommendation

For sharing with developers, the preferred artifact is a Git format-patch series based on 06f3bc759bf5 (sdm670-v7.0.10), not only a prebuilt boot image. The patch series preserves Signed-off-by metadata, reviewable commit boundaries, and the exact source changes.

```
BASE=06f3bc759bf5
mkdir -p /tmp/bonito-pm660-patches
git format-patch -o /tmp/bonito-pm660-patches "$BASE"..HEAD

# Optional single combined diff for convenience:
git diff --binary "$BASE"..HEAD > /tmp/bonito-pm660-usbc-final.patch
```

End of engineering validation report